SkillGraph Nexus

AI-Adaptive Skill Gap Analysis and Personalized Learning Path Engine

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Project Title: SkillGraph Nexus
Subtitle: AI-Adaptive Onboarding and Skill Gap Mapping Platform

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1. Problem Understanding

Modern corporate onboarding and professional learning systems often follow a **static, one-size-fits-all training model**.

This approach introduces several inefficiencies:

• Experienced professionals waste time revisiting concepts they already know.
• Beginners often struggle because advanced modules assume prior knowledge.
• Organizations lack visibility into the **actual skill readiness of candidates or employees**.
• Learning recommendations are generic and not personalized.

As hiring and career paths become increasingly interdisciplinary, the gap between **existing skills and job requirements** continues to widen.

Traditional learning platforms recommend courses without evaluating:

• the learner's current knowledge
• the target role requirements
• the optimal learning path to close the skill gap.

Objective

The objective of this project is to build an **AI-Adaptive Learning Engine** capable of:

1. Parsing a user's **resume or skill profile**
2. Understanding the **requirements of a target job role**
3. Performing **intelligent skill gap analysis**

4. Generating a **personalized learning pathway** to bridge the gap efficiently.

The solution aims to reduce redundant training while ensuring learners gain the **exact competencies required for their desired role**.

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2. Proposed Solution

SkillGraph Nexus is an AI-powered platform that analyzes a user's resume and compares it with job role requirements to generate **adaptive learning roadmaps**.

The platform performs the following steps:

1. Resume ingestion and parsing
2. Skill extraction using AI
3. Job description requirement analysis
4. Skill comparison and gap identification
5. Personalized learning recommendations

Unlike traditional systems, the platform dynamically updates recommendations when the user changes the **target role**, demonstrating adaptability across domains.

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3. System Architecture

SkillGraph Nexus follows a **modular architecture combining frontend UI, AI processing, and backend infrastructure**.

The system architecture consists of three main layers:

1. User Interface Layer

Provides the interactive web application where users:

• Upload resumes
• Enter target job roles
• View skill gap analysis
• Track progress and recommendations

2. AI Processing Layer

Responsible for:

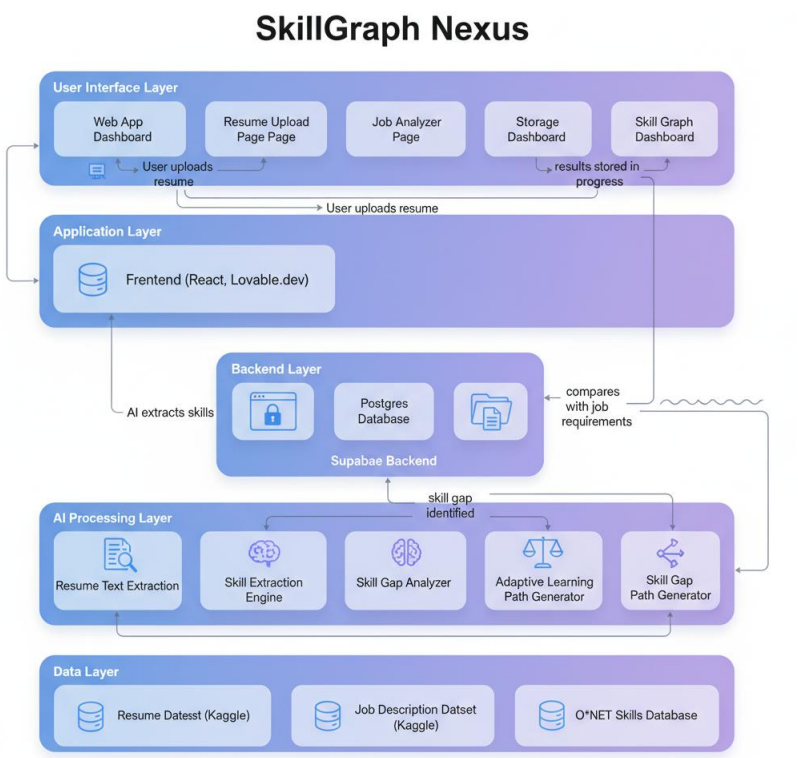
• Resume text parsing
• Skill extraction

• Job requirement analysis
• Skill gap detection

3. Backend & Data Layer

Manages:

• user authentication
• data storage
• skill tracking
• persistent analysis results



Label: System Architecture of SkillGraph Nexus

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4. System Workflow

The system workflow demonstrates how user inputs are transformed into personalized skill recommendations.

Step 1 – Resume Upload

The user uploads their resume through the web interface.

Step 2 – Resume Parsing

The system extracts relevant textual information such as:

- education
- technical skills
- experience
- projects

Step 3 – Skill Extraction

AI models identify and categorize skills present in the resume.

Step 4 – Job Requirement Analysis

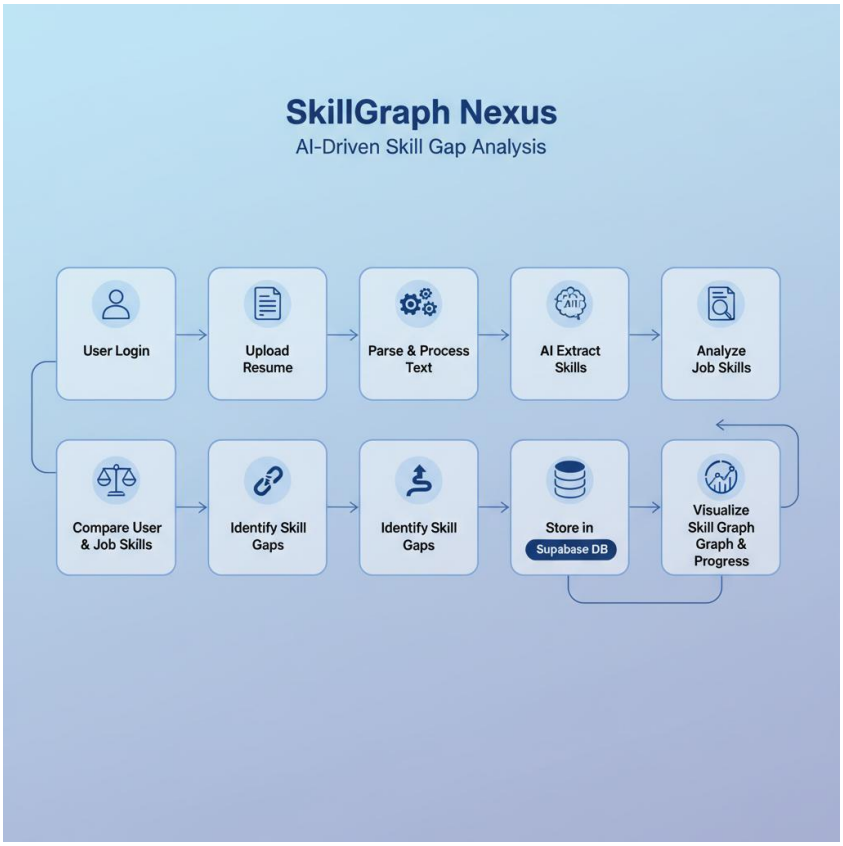
The user specifies a target job role, and the system retrieves the typical skills required for that role.

Step 5 – Skill Gap Analysis

The system compares extracted skills against required skills and identifies missing competencies.

Step 6 – Learning Path Generation

A personalized roadmap is generated with recommended skills and learning timelines.



Label: End-to-End Workflow of SkillGraph Nexus

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5. Technology Stack Used

SkillGraph Nexus was built using modern AI and web technologies to ensure scalability and rapid prototyping.

Frontend

- React-based web interface
- Interactive dashboards
- Skill graph visualizations

Backend

- Serverless backend architecture
- RESTful APIs

Database & Authentication

The platform uses Supabase for:

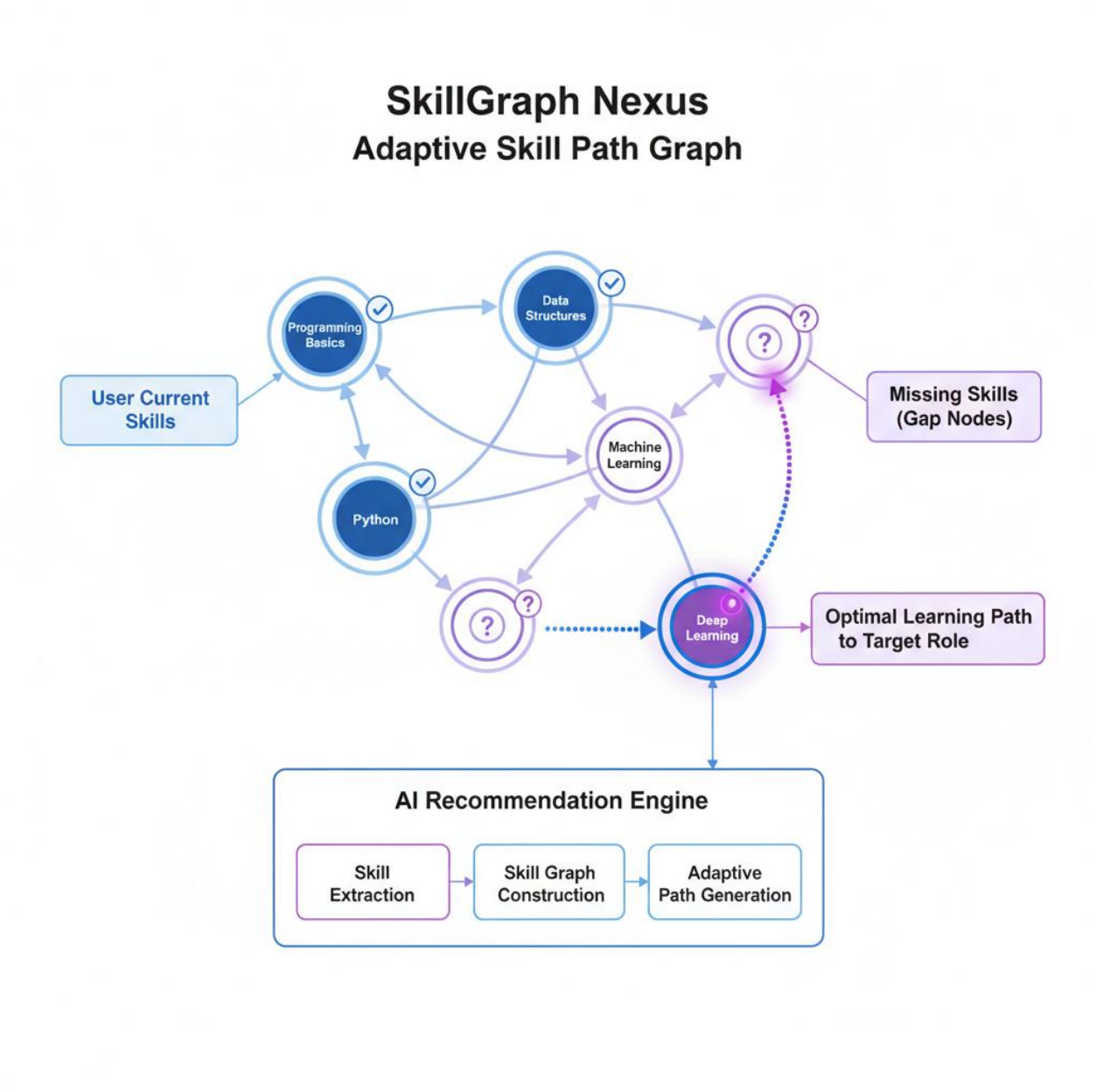
- user authentication
- secure data storage
- skill tracking and leaderboard management

AI & NLP Components

AI-driven modules perform:

- resume parsing
- skill extraction
- job description analysis
- skill comparison and gap detection

These components enable the system to interpret unstructured resume data and convert it into structured skill profiles.



Label: AI Skill Graph Mapping and Skill Gap Detection

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6. Adaptive Learning Path Algorithm

The core intelligence of SkillGraph Nexus lies in its **Adaptive Skill Mapping Engine**.

The algorithm operates in three stages:

Stage 1 – Skill Extraction

Skills are extracted from the resume using natural language processing techniques.

Stage 2 – Skill Matching

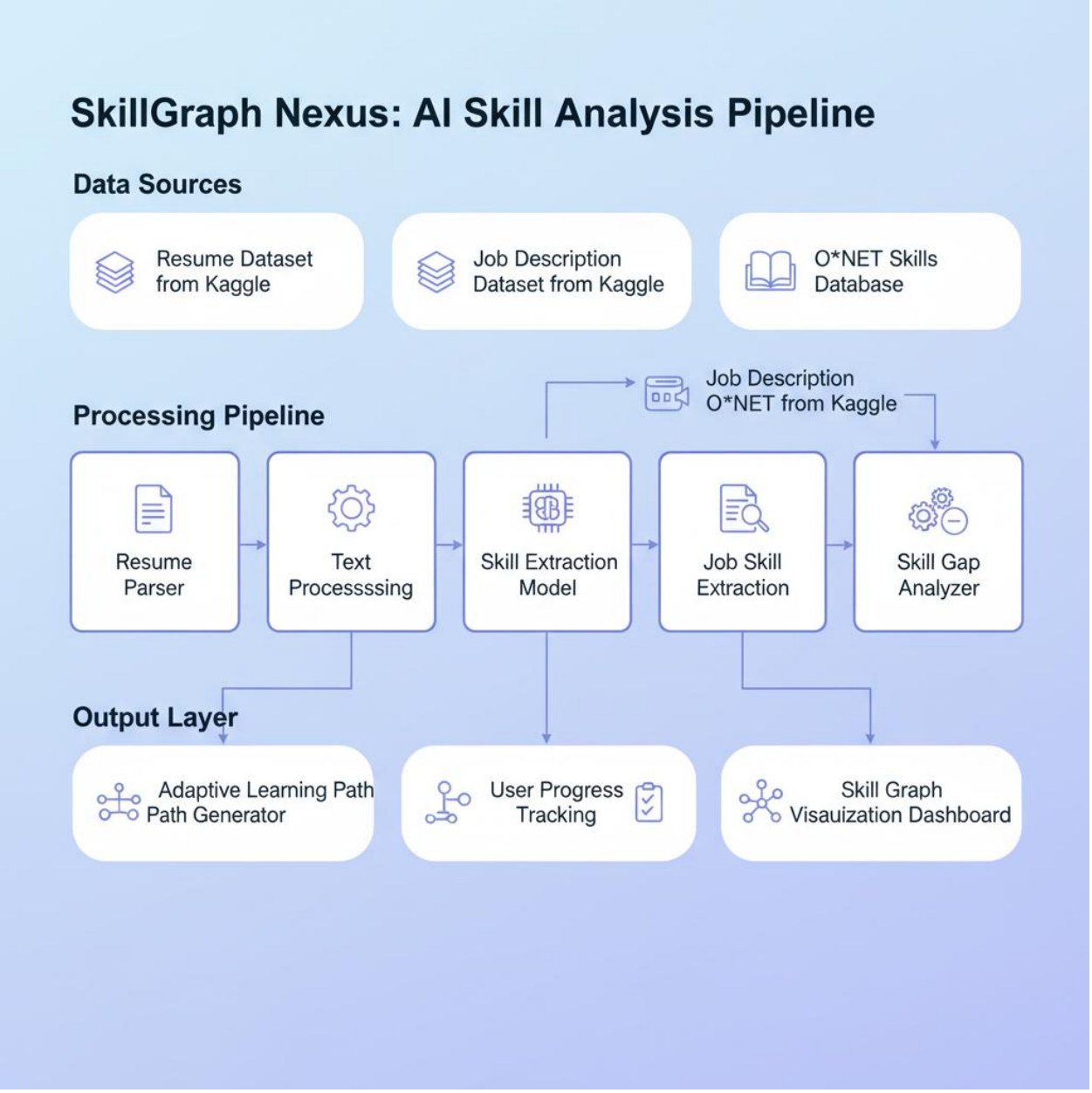
Extracted skills are compared against required job skills using similarity matching and classification.

Stage 3 – Adaptive Path Generation

The system generates a learning roadmap prioritizing skills based on:

- importance for the target role
- estimated learning time
- user's existing competencies

This allows the system to recommend **efficient learning pathways instead of generic training programs**.



Label: AI Adaptive Skill Path Generation Algorithm

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7. Key Features of the Solution

SkillGraph Nexus introduces several innovative features designed to improve onboarding and learning efficiency.

AI Resume Analysis

Automatically parses resumes and extracts relevant skills and experience.

Dynamic Skill Gap Detection

Identifies missing competencies required for a specific job role.

Personalized Learning Path

Generates skill-based learning recommendations with estimated completion timelines.

Cross-Domain Adaptability

The system dynamically adapts when users switch between job roles.

Example:

Resume analyzed for:

- University Physics Teacher
- Web Development Mentor

Each role generates a **different skill gap profile**, demonstrating adaptability.

Skill Graph Visualization

Users can visually understand their skill landscape through an interactive graph.

Secure Data Management

User data and analysis results are securely stored using backend database services.

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8. Impact and Potential Applications

SkillGraph Nexus can significantly improve:

Corporate Onboarding

Organizations can tailor training programs to individual employee skill levels.

Career Planning

Students and professionals can identify missing skills required for their desired roles.

Workforce Upskilling

Companies can efficiently reskill employees based on evolving industry requirements.

9. Future Improvements

Future versions of the platform could include:

- advanced knowledge tracing models
- integration with real course providers
- automated resume optimization
- progress tracking and learning streaks
- gamified career development features

These enhancements would further strengthen the system as a **comprehensive AI career development platform**.

10. Conclusion

SkillGraph Nexus demonstrates how AI can transform traditional onboarding and skill development processes.

By combining resume analysis, job requirement mapping, and adaptive learning path generation, the platform enables users to clearly understand:

- where they currently stand
- what skills they need
- how to efficiently reach their career goals.

This approach reduces redundant training and ensures learners focus only on the skills that truly matter for their target role.